Ramen TTA – Experiment Report

Auto-generated by experiments.py

2026-08-31 16:58:49

1 Run configuration

- **date**: 2026-08-31 16:58:49
- **sweep_run**: False
- **diagnostics_run**: False
- **tsne_run**: False
- **device**: cuda
- **num_result_rows**: 285
- **datasets**: CIFAR100C, CIFAR10C

2 Average accuracy per algorithm / flag

Each cell is the mean over all corruption domains of that run's per-domain accuracy (%). Compare adaptation on/off (NoAdapt vs Tent vs Ramen) and the `use_tbn` toggle.

dataset	tta_algo	off	on
CIFAR100C	NoAdapt	37.10	–
CIFAR100C	Ramen	46.12	46.12
CIFAR100C	Tent	41.14	41.14
CIFAR10C	NoAdapt	61.36	–
CIFAR10C	Ramen	72.63	72.63
CIFAR10C	Tent	66.71	66.71

Table 1: Mean per-domain accuracy (%) by dataset, algorithm and use_tbn.

dataset	tta_algo	use_tbn	avg_acc
CIFAR100C	Ramen	False	46.12
CIFAR100C	Ramen	True	46.12
CIFAR100C	Tent	False	41.14
CIFAR100C	Tent	True	41.14
CIFAR100C	NoAdapt	False	37.10
CIFAR10C	Ramen	False	72.63
CIFAR10C	Ramen	True	72.63
CIFAR10C	Tent	False	66.71
CIFAR10C	Tent	True	66.71
CIFAR10C	NoAdapt	False	61.36

Table 2: All runs ranked by average accuracy within each dataset.

3 Per-domain accuracy

CIFAR100C

domain	NoAdapt	Ramen	Tent
glass_blur	20.16	29.34	23.67
gaussian_noise	22.83	30.50	24.87
shot_noise	24.42	32.77	27.03
pixelate	26.29	40.42	31.53
elastic_transform	29.21	36.83	31.81
impulse_noise	29.58	42.80	34.49
jpeg_compression	33.57	39.15	37.24
contrast	35.27	51.18	43.93
fog	41.77	52.65	45.27
motion_blur	43.60	52.20	46.98
defocus_blur	43.56	52.46	49.52
frost	50.48	54.72	50.98
snow	48.91	55.23	52.63
zoom_blur	48.64	55.65	53.95
brightness	58.15	65.96	63.24

Table 3: CIFAR100C: per-domain accuracy (%) by algorithm (use_tbn=off), sorted ascending.

domain	NoAdapt	Ramen	Tent
glass_blur	20.16	29.34	23.67
gaussian_noise	22.83	30.50	24.87
shot_noise	24.42	32.77	27.03
pixelate	26.29	40.42	31.53
elastic_transform	29.21	36.83	31.81

Table 4: CIFAR100C: 5 weakest corruption domains (by Tent).

CIFAR10C

domain	NoAdapt	Ramen	Tent
gaussian_noise	38.66	60.84	38.60
shot_noise	42.59	63.35	41.56
impulse_noise	42.75	56.13	45.53
glass_blur	43.91	64.11	56.22
pixelate	51.88	65.18	60.55
jpeg_compression	58.58	67.32	63.86
elastic_transform	61.97	71.95	69.57
motion_blur	66.68	77.26	72.63
fog	68.85	78.38	73.94
contrast	67.07	79.68	75.54
defocus_blur	72.61	77.07	78.05
zoom_blur	71.69	78.84	78.18
snow	73.78	80.79	79.68
frost	75.69	80.42	79.85
brightness	83.62	88.14	86.84

Table 5: CIFAR10C: per-domain accuracy (%) by algorithm (use_tbn=off), sorted ascending.

domain	NoAdapt	Ramen	Tent
gaussian_noise	38.66	60.84	38.60
shot_noise	42.59	63.35	41.56
impulse_noise	42.75	56.13	45.53
glass_blur	43.91	64.11	56.22
pixelate	51.88	65.18	60.55

Table 6: CIFAR10C: 5 weakest corruption domains (by Tent).

4 Per-domain accuracy (figure)

Bars sorted ascending; dashed line = that run's own average; red bars are domains below the average (dragging the total down).

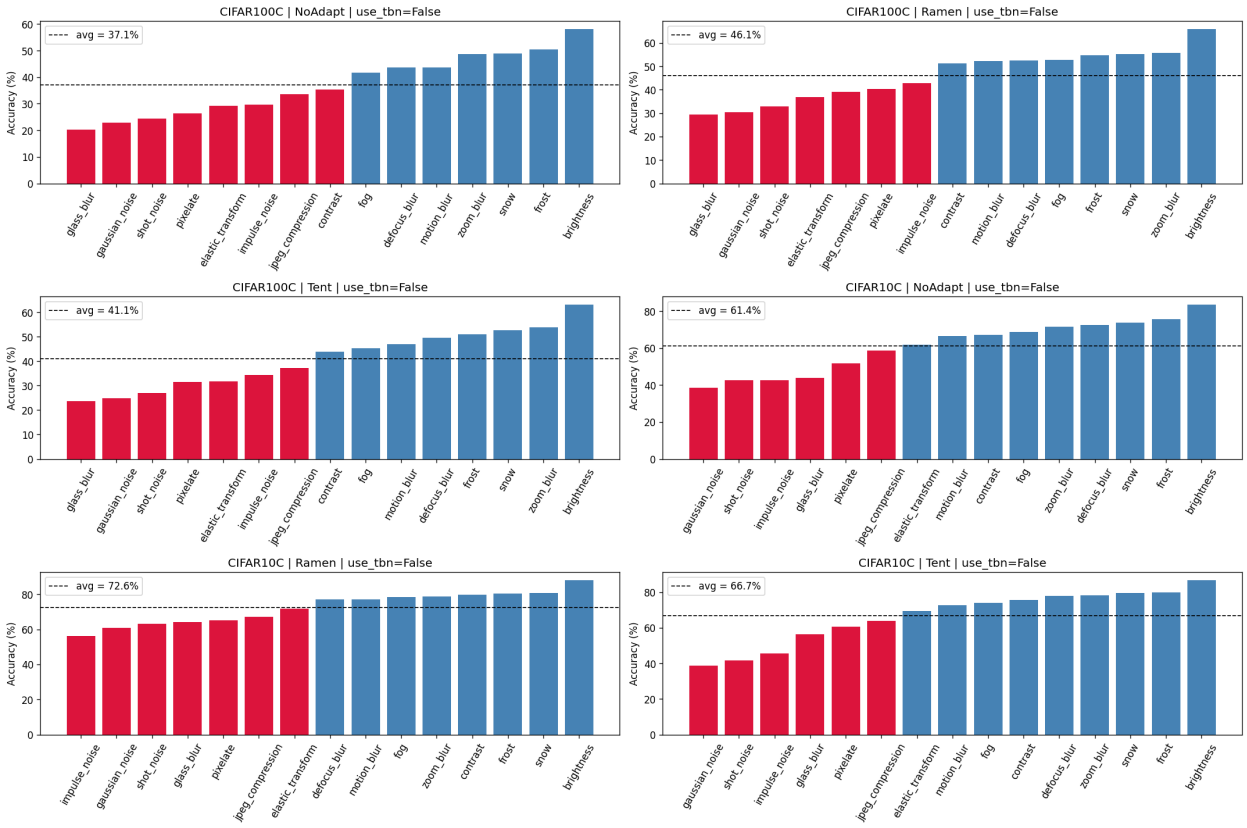


Figure 1: Per-domain accuracy, per (dataset, algorithm).